

- (A) Let $f : \Omega \rightarrow \mathbb{R}$ be a $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable *nonnegative* function. Define for $n \geq 1$, and $x \in \Omega$

$$f_n(x) = \begin{cases} \frac{(i-1)}{2^n}, & \text{if } \frac{(i-1)}{2^n} \leq f(x) < \frac{i}{2^n} \text{ for } i = 1, \dots, n2^n \\ n, & \text{if } f(x) \geq n \end{cases}$$

Then prove the following:

- (i) For each $n \geq 1$, $f_n(x)$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable.
 - (ii) For each $x \in \Omega$, $f_n(x)$ is increasing with n (i.e $\{f_n(\cdot)\}$ is an increasing seq. of fns).
 - (iii) Prove that $f_n(x) \rightarrow f(x)$ pointwise (i.e for each fixed x), as $n \rightarrow \infty$.
- (B) If $f, g \in L_1(\Omega, \mathcal{F}, \mu)$,

show

$$\min(f, g) \in L_1(\Omega, \mathcal{F}, \mu)$$

and

$$\min \left(\int f d\mu, \int g d\mu \right) \geq \int \min(f, g) d\mu.$$